

MED 610 Examination

Clinical Reasoning and Differential Diagnosis

Instructions:

- Answer all questions.
 - For Questions 1–4, choose the best option.
 - For Questions 5–8, write brief but precise answers.
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1. A 55-year-old male presents with sudden onset severe chest pain radiating to the back, described as “tearing.” Blood pressure is 180/110 mmHg in the right arm and 140/90 mmHg in the left arm. What is the most likely diagnosis?

(A) Acute myocardial infarction

(B) Aortic dissection

(C) Pulmonary embolism

(D) Pericarditis

 2. A 25-year-old female presents with polyuria, polydipsia, and weight loss. Random blood glucose is 350 mg/dL. Urinalysis shows glucose and ketones. What is the most appropriate initial diagnosis?

(A) Type 2 diabetes mellitus

(B) Diabetic ketoacidosis

(C) Hyperosmolar hyperglycemic state

(D) Diabetes insipidus

 3. A 40-year-old male presents with jaundice, dark urine, and pale stools. Laboratory tests show elevated conjugated bilirubin and alkaline phosphatase. What type of jaundice is most likely?

(A) Hemolytic jaundice (prehepatic)

(B) Hepatocellular jaundice (hepatic)

(C) Obstructive jaundice (posthepatic)

(D) Physiologic jaundice

 4. A 60-year-old smoker presents with chronic cough, hemoptysis, and weight loss. Chest X-ray shows a centrally located mass with hilar lymphadenopathy. Which type of lung cancer is most likely?

(A) Adenocarcinoma

- (B) Squamous cell carcinoma
 - (C) Small cell carcinoma
 - (D) Bronchioloalveolar carcinoma
5. Describe a systematic approach to developing a differential diagnosis for a patient presenting with acute abdominal pain. Include consideration of anatomical location, character of pain, associated symptoms, and initial diagnostic tests.
 6. A 70-year-old male presents with progressive dyspnea, orthopnea, and bilateral leg edema. Jugular venous distension is noted. Construct a differential diagnosis for heart failure, distinguishing between left-sided, right-sided, and biventricular failure, and describe the key clinical features of each.
 7. Explain the diagnostic approach to a patient presenting with fever of unknown origin (FUO). What are the major categories of causes, and what investigations would you prioritize in the initial workup?
 8. Discuss the differential diagnosis of altered mental status in an elderly patient. What are the key differences between delirium and dementia, and what acute medical conditions should be urgently considered and ruled out?